TRANSITIVITY OF THE UNIVERSAL THICKENING AND THE CONORMAL SEQUENCE FOR TWO FORMALLY UNRAMIFIED MORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1437-1455 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 1457-1479, the complete author proof (21 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Universal first order thickenings of formally unramified morphisms, conormal sheaves and their functoriality; the comparison is retained.

Source reference. Original lemma lemma-universal-thickening-functorial: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 1220–1242. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma morphisms-lemma-conormal-functorial-flat: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 1479–1479. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma morphisms-lemma-transitivity-conormal: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 1479–1479. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $Z \to Y \to X$ be formally unramified morphisms of schemes.*

(1) *If $Z \subset Z'$ is the universal first order thickening of Z over X and $Y \subset Y'$ is the universal first order thickening of Y over X, then there is a morphism $Z' \to Y'$ and $Y \times_{Y'} Z'$ is the universal first order thickening of Z over Y.*

(2) *There is a canonical exact sequence*

$$i^*\mathcal{C}_{Y/X} \to \mathcal{C}_{Z/X} \to \mathcal{C}_{Z/Y} \to 0$$

where the maps come from Lemma lemma-universal-thickening-functorial and $i : Z \to Y$ is the first morphism.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. The map $h : Z' \to Y'$ in (1) comes from Lemma lemma-universal-thickening-functorial. The assertion that $Y \times_{Y'} Z'$ is the universal first order thickening of Z over Y is clear from the universal properties of Z' and Y'. By Morphisms, Lemma morphisms-lemma-transitivity-conormal we have an exact sequence

$$(i')^*\mathcal{C}_{Y\times_{Y'}Z'/Z'} \to \mathcal{C}_{Z/Z'} \to \mathcal{C}_{Z/Y\times_{Y'}Z'} \to 0$$

where $i' : Z \to Y \times_{Y'} Z'$ is the given morphism. By Morphisms, Lemma morphisms-lemma-conormal-functorial-flat there exists a surjection $h^*\mathcal{C}_{Y/Y'} \to \mathcal{C}_{Y\times_{Y'}Z'/Z'}$. Combined with the equalities $\mathcal{C}_{Y/Y'} = \mathcal{C}_{Y/X}$, $\mathcal{C}_{Z/Z'} = \mathcal{C}_{Z/X}$, and $\mathcal{C}_{Z/Y\times_{Y'}Z'} = \mathcal{C}_{Z/Y}$ this proves the lemma. □